Calculating ionisation energies (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Explain the trends and discontinuities in first ionisation energy (IE) across a period and down a group.
- Calculate the value of the first IE from spectral data that gives the wavelength or frequency of the convergence limit.

What type of forces hold an atom together? So far you have learned that an atom contains three sub-atomic particles – the proton, neutron and electron. The protons and neutrons are located in the nucleus and are held together by the strong nuclear force. This strong force is required to overcome the repulsion between the positively charged protons – without it the nucleus would blow itself apart. Electrons are held in position by the electrostatic attraction between the positively charged protons and negatively charged electrons. If enough energy is supplied, this attraction can be overcome and electrons can be removed from the atom. In this section you will learn how to calculate this energy, known as the ionisation energy.

Making connections

Ionisation energy is covered in more detail in section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/). The periodic table places elements in rows and columns, organising the elements based on similarities. There are 18 vertical columns, known as groups, and 7 horizontal rows, known as periods. Ionisation energy decreases for elements that are in the same group and increases for elements in the same period.

Trends in ionisation energy

Ionisation refers to the process of removing electrons from an atom in its ground state. For hydrogen this involves an electron transition from $n = 1$ to $n = \infty$ (**Figure 1**). Recall from section S1.3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/) that an electron in the $n = \infty$ energy level is said to be removed from the attraction of the nucleus and the atom has been ionised.

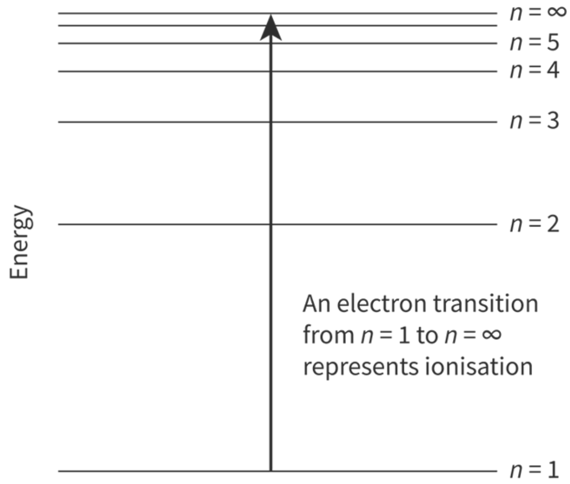

Figure 1. When an electron in a hydrogen atom transitions from n = 1 to n = ∞, it is said to be ionised.

More information for figure 1

The diagram illustrates an electron transitioning within a hydrogen atom from the n=1 orbit (ground state) to n=∞ (ionized state). It uses concentric circles to represent the atomic orbits, with labels indicating the n=1, n=2, n=3, and n=∞ energy levels. An arrow points from the n=1 level to n=∞, indicating the ionization process where the electron moves from being tightly bound to the nucleus to being completely free. Additional labels may include terms like 'ground state' and 'ionized hydrogen atom.'

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To remove electrons from an atom, energy is required to overcome the electrostatic attraction between the positively charged nucleus and the negatively charged electrons. The strength of this attraction depends on two factors:

1. the number of protons in the nucleus, which is known as the nuclear charge.
2. the distance between the nucleus and the outer electrons, which is known as the atomic radius.

The general trend in ionisation energy is shown in the graph in **Figure 2**. This is covered in more detail in section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/).

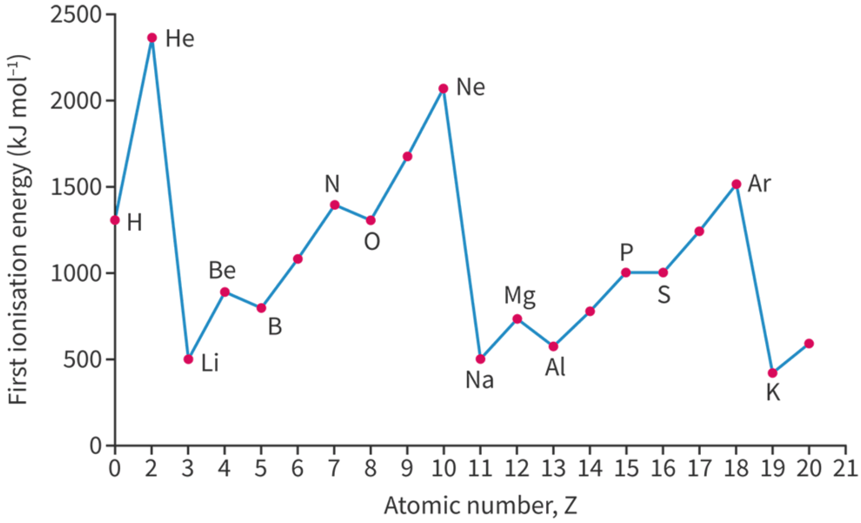

Figure 2. The trend in ionisation for the first 20 elements in the periodic table.

More information for figure 2

The graph illustrates the trend in ionization energies for the first 20 elements of the periodic table. The X-axis represents the atomic number, going from 1 (Hydrogen) to 20 (Calcium). The Y-axis indicates the ionization energy in electron volts (eV), with labeled intervals. The graph showcases two series of peaks and troughs. Notable features include:

- A sharp increase in ionization energy from Hydrogen to Helium.
- A dip at Lithium, followed by a rise to Beryllium and another dip at Boron.
- Steady increases and periodic dips are seen again at elements Nitrogen and Oxygen, highlighting the discontinuities between certain elements (e.g., Be to B and N to O).
- For period 3, a rise from Sodium, with dips and rises similar to the pattern in period 2.
- The graph highlights elements with higher ionization energies and discontinuities that are less pronounced between elements 11 to 18 (Sodium to Argon), showing similar patterns to the previous period but shifted to higher values on the Y-axis.
- Concludes with a discussion of periodic trends and exceptions to these trends in the context of periods.

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Looking at **Figure 2**, what overall trend can you observe for elements 3 to 10 (lithium to neon)? Are there any elements that don't seem to follow the trend? What about for elements 11 to 18 (sodium to argon)? Is the general trend the same? Once again, are there any elements that don't seem to follow the trend? Close inspection of this graph reveals two discontinuities in each period – between Be and B and also N and O in period 2 and between Mg and Al and also P and S in period 3.

The decrease in first ionisation energy from beryllium to boron can be explained by examining their electron configurations:

- beryllium has the electron configuration $1s^2\ 2s^2$
- boron has the electron configuration $1s^2\ 2s^2\ 2p^1$

Electrons in p orbitals are of higher energy and further from the nucleus than electrons in s orbitals, therefore they require less energy to remove (**Figure 3**).

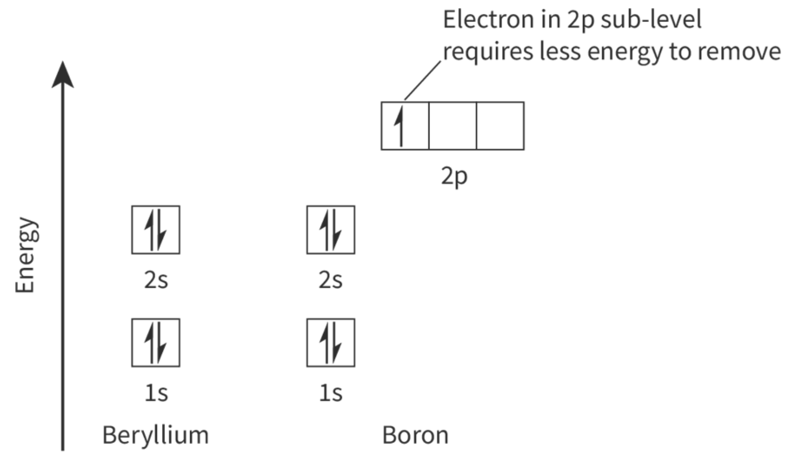

Figure 3. The orbital diagrams for beryllium and boron.

More information for figure 3

This image displays orbital diagrams for beryllium and boron, comparing their energy levels. On the vertical axis, energy is labeled, indicating increasing energy as you move upwards. The diagram for beryllium shows two energy levels, labeled as 1s and 2s, each with a filled orbital represented by arrows indicating electron spin within a rectangular box.

Next to beryllium, the diagram for boron also shows the 1s and 2s levels with filled orbitals. Above these, a 2p level is shown with one single electron indicated by an arrow in a partially filled rectangular box. An annotation reads, "Electron in 2p sub-level requires less energy to remove," emphasizing that the 2p orbital's electron is of higher energy, thus easier to remove than those in s orbitals.

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The same explanation can be applied for the decrease in ionisation energy from magnesium to aluminium, except that the electron configurations are $1s^2\ 2s^2\ 2p^6\ 3s^2$ and $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$ respectively. The electron removed from aluminium is located in a 3p orbital which is further from the nucleus and requires less energy to remove than the electron removed from the 3s orbital in magnesium.

To explain the decrease in ionisation energy between nitrogen and oxygen we will examine their electron configurations:

- nitrogen has the electron configuration $1s^2\ 2s^2\ 2p^3$
- oxygen has the electron configuration $1s^2\ 2s^2\ 2p^4$

Figure 4 shows the electron in box diagrams for nitrogen and oxygen.

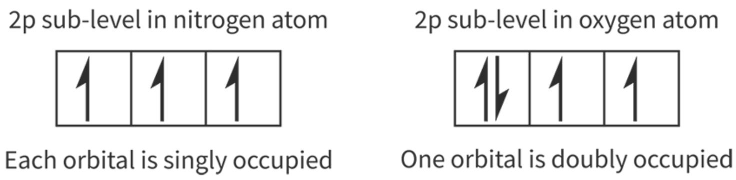

Figure 4. The orbital diagrams for nitrogen and oxygen.

More information for figure 4

The image depicts orbital diagrams for nitrogen and oxygen. In the nitrogen diagram, there are three boxes representing p orbitals, each containing a single upward arrow, indicating singly occupied orbitals. In contrast, the oxygen diagram shows three boxes as well, but one of the boxes contains two arrows (one up and one down), indicating a doubly occupied orbital. This illustrates how nitrogen has three singly occupied p orbitals, while oxygen has one doubly occupied orbital, aligning with the description that an electron in the doubly occupied orbital of oxygen requires less energy to remove due to repulsion from the second electron.

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Nitrogen has three singly occupied p orbitals but in oxygen, one orbital is doubly occupied. An electron in a doubly occupied orbital is repelled by the second electron and requires less energy to remove than an electron in a half-filled orbital. The same explanation applies for the decrease in ionisation energy from phosphorus to sulfur, except that the electrons are being removed from the 3p sublevel.

Calculating ionisation energy

In section S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/), you learned that energy levels converge at high energy. The point at which the energy levels in an atom converge is known as the convergence limit. At this point the electron has been removed from the attraction of the nucleus and the atom has been ionised. If the wavelength or frequency of the convergence limit is known, the ionisation energy can be calculated. To calculate the ionisation energy, we will make use of the equations and constants in **Tables 1** and **2**.

Table 1. Equation for energy.

Equation	Symbols	Units
$E = hf$	E is energy	Joules (J)
	h is Planck's constant, 6.63×10^{-34}	Joule seconds (J s)
	f is the frequency	seconds^{-1} (s^{-1})

Table 2. Equation for speed of light.

Equation	Symbols	Units
$c = \lambda f$	c is the speed of light 3.00×10^{8}	metres per second (m s^{-1})
	λ is wavelength	metres (m)

Equation	Symbols	Units
	f is the frequency	seconds^{-1} (s^{-1})

Study skills

The equations and constants listed above can be found in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) and section 2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) of the DP chemistry data booklet.

Worked example 1

The frequency of the convergence limit for the hydrogen atom is $3.28 \times 10^{15}\ \text{s}^{-1}$.

Calculate the ionisation energy for hydrogen in kJ mol^{-1}.

In the question the frequency of convergence limit is given so we can use the equation:

$$\begin{aligned} E &= hf \\ &= 6.63 \times 10^{-34} \times 3.28 \times 10^{15} \\ &= 2.17 \times 10^{-18}\ \text{J} \end{aligned}$$

This is the ionisation energy in joules (J) for a single atom of hydrogen. To calculate the ionisation energy in kJ mol^{-1}, we need to multiply by the Avogadro constant and divide by 1000 to convert to kJ.

$$\begin{aligned} E &= \frac{2.17 \times 10^{-18}\ \times 6.02 \times 10^{23}}{1000} \\ &= 1310\ \text{kJ mol}^{-1}\ (3\ \text{s.f.}) \end{aligned}$$

Worked example 2

Determine the wavelength of a photon that will cause the first ionisation of copper. The ionisation energy of copper is 745 kJ mol^{-1}.

In the question we will calculate the wavelength of a single photon of energy required to ionise a copper atom.

First, we will find the energy of a photon in joules (J) to ionisation a single copper atom:

$$\begin{aligned} E &= \frac{745\ \times 1000}{6.02 \times 10^{23}} \\ &= 1.24 \times 10^{-18}\ \text{J}\ (3\ \text{s.f.}) \end{aligned}$$

Next, we will calculate the frequency of the photon:

$$E = hf$$
$$1.24 \times 10^{-18} = 6.63 \times 10^{-34} \times f$$
$$f = 1.87 \times 10^{15}\ \mathrm{s^{-1}}\ (3\ \mathrm{s.f.})$$

Finally, we will calculate the wavelength of the photon:

$$c = \lambda f$$
$$3.00 \times 10^{8} = 1.87 \times 10^{15} \times \lambda$$
$$\lambda = 1.60 \times 10^{-7}\ \mathrm{m}\ (3\ \mathrm{s.f.})$$

Theory of Knowledge

Observing colour with our sense of sight is something not all people can do, and many ‘colours’ fall out of the visible spectrum. To what extent do we put faith in technology to ‘see’ what we cannot see ourselves?

Making connections

How does the trend in ionisation energy values across a period and down a group explain the trends in properties of metals and non-metals? In subtopic S3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/), you will learn about the trend in ionisation energy can be used to distinguish between metal and non-metal elements.

In the following activity, you will practise how to calculate the ionisation energy using a database.

Activity

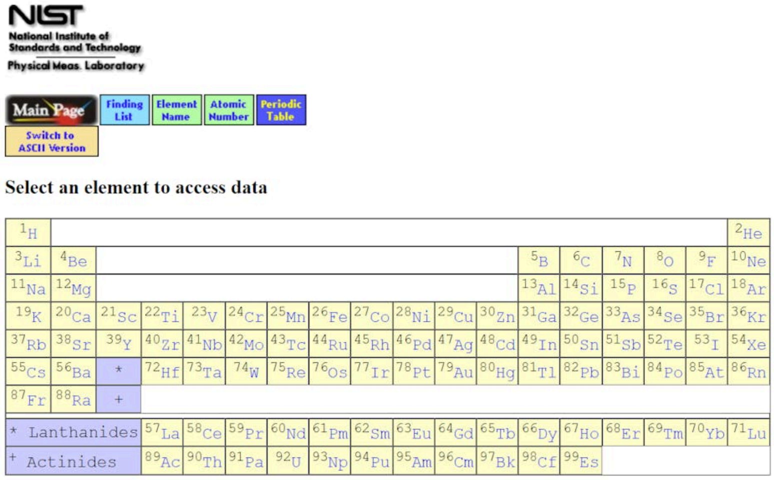
- **IB learner profile attribute**: Inquirers
- **Approaches to learning**: Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

In this activity, you will use a database to calculate the ionisation energy from the convergence limit of an atom.

1. Start by clicking on this link.

   http://physics.nist.gov/PhysRefData/Handbook/periodictable.htm (http://physics.nist.gov/PhysRefData/Handbook/periodictable.htm)

   You should see this periodic table below. Choose an element from the periodic table and click on it. In this example, we will choose carbon.



More information

The image shows a periodic table from NIST (National Institute of Standards and Technology). The title suggests selecting an element to access data. The table consists of various chemical elements arranged by atomic number, symbol, and name. Key sections include groups like Lanthanides and Actinides, displayed in separate rows below the main table. Elements are arranged in columns and rows representing groups and periods. Options like "Main Page," "Finding List," "Element Name," and "Atomic Number" are located at the top along with a button to switch to an ASCII version of the table.

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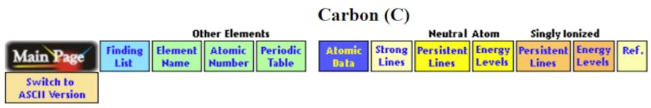
1. Click on ‘Neutral atom’ and ‘Energy levels’.



More information

This image is a screenshot of a webpage header featuring navigation buttons related to the element Carbon (C). It has multiple labeled sections. Starting from the left, there is a button labeled 'Main Page' with an option below to 'Switch to ASCII Version.' Next, there are labeled sections titled 'Other Elements' and 'Neutral Atom.' Under 'Other Elements,' the tabs include 'Finding List,' 'Element Name,' 'Atomic Number,' and 'Periodic Table,' each with its own distinct color background. Under 'Neutral Atom,' the tabs are 'Atomic Data,' 'Strong Lines,' 'Persistent Lines,' and 'Energy Levels.' The section labeled 'Singly Ionized' includes 'Persistent Lines,' 'Energy Levels,' and 'Ref.' Each tab is presented in a colored box, indicating distinct navigational links.

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1. Scroll to the bottom of the page where it says ‘C I Ground State’. Make a note of the value for the ionisation energy.

2. This value is the wavenumber (in cm^{-1}) of the convergence limit for a carbon atom. To convert to wavelength in metres, we need to take the reciprocal of the wavenumber and divide by 100. So for carbon:

$$\begin{aligned}\lambda &= \frac{\frac{1}{90\,820}}{100}\\ &= 1.10 \times 10^{-7}\ \mathrm{m}\ (3\ \mathrm{s.f.})\end{aligned}$$

3. Now that we have the wavelength of the convergence limit of a carbon atom, we can calculate the frequency of the convergence limit.

$$\begin{aligned}c &= \lambda f\\ 3.00 \times 10^{8} &= 1.10 \times 10^{-7} \times f\\ f &= 2.73 \times 10^{15}\ \mathrm{s}^{-1}\end{aligned}$$

4. Next, we will calculate the ionisation energy in kJ mol^{-1}.

$$\begin{aligned}E &= hf\\ &= 6.63 \times 10^{-34} \times 2.73 \times 10^{15}\ \mathrm{s}^{-1}\\ &= 1.81 \times 10^{-18}\ \mathrm{J}\end{aligned}$$

This is the ionisation energy in joules (J) for a single atom of carbon. To convert to kJ mol^{-1}, we need to multiply by the Avogadro constant and divide by 1000.

$$\frac{1.81 \times 10^{-18} \times 6.02 \times 10^{23}}{1000} = 1090\ \mathrm{kJ\ mol^{-1}}$$

If we compare this to the value in section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet, which is 1086 kJ mol^{-1}, we can see our calculated value is accurate to three significant figures.

Next, choose a different atom from the periodic table and calculate the ionisation energy in kJ mol^{-1}. Once you have calculated a value, check it with section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet.